

Sachin Mishra

Bengaluru, IN | +91 9111113267 | sachin.mishra@iiitb.ac.in | https://imsachin010.github.io/Sachin_Portfolio/ |
[LinkedIn: sachinmishra010](https://www.linkedin.com/in/sachinmishra010) | <https://scholar.google.com/citations?user=acsumE0AAAAJ>

RESEARCH INTERESTS

AI Alignment, LLM Evaluations, and Resource-Aware ML Systems. My research focuses on epistemic reasoning in multi-source conflict resolution and building instrumented benchmarks to measure the accuracy-efficiency trade-offs of foundation models. My current MS thesis explores multi-objective optimization and Pareto-frontier analysis to ensure AI systems remain reliable, grounded, and computationally sustainable at scale.

EDUCATION

International Institute of Information Technology, Bangalore (IIIT-B) Bangalore, India
MS Research in Artificial Intelligence & Data Science; GPA: 3.63/4; Advisor: Prof. Aswin Kannan 2025 – 2027
Coursework: Machine Learning, Optimization, Reinforcement Learning, Self-Supervised Learning, Theory of LLMs, Mathematics for ML, Deep Learning & Foundational Models.

Government Engineering College (GEC) Raipur Raipur, India
Bachelor of Technology in Computer Science & Engineering; GPA: 3.58/4 2021 – 2025

PUBLICATIONS

- **When Sources Disagree: How Language Models Decide What to Trust.** AAI 2027 (Under Review)
- **Range: Resource-Aware Non-monotonicity in Generation Efficiency.** IKDD CODS (Under Review)
- **Do Vision-Language Models Actually See?** TMLR (Under Review) [Paper]
- **Reliability-Aware Structured Synthetic Data Generation via Schema Enforcement and Layered Repair.** SynthAI @ SIGMOD 2026 [Paper]
- **Automated Assessment of Rooftop Solar Energy Potential Using Deep Learning-Driven Building Footprint Extraction.** ADCOM 2025 [Paper]
- **Dynamic Query Handling with RAG Fusion for PDF-Based Knowledge Retrieval.** OTCN 2025 [Paper]
- **TransDFd: A Deepfake Detection System at Mesoscopic Level.** IEEE IATMSI 2025 [Paper]
- **IoT ML Driven Holistic Health Monitoring and Fitness Assessment.** IEEE SCES 2024 [Paper]
- **UDR Fused Multimodal Approach for Disease Classification in Large Scale Dataset.** Springer LNNS 2023 [Paper]
- **AI-Enabled Audio Processing Pen for Transcription.** IN Design Patent 189,584 [Patent]

SELECTED RESEARCH PROJECTS

LLM Epistemic Reasoning & Alignment

- **Reliable Multi-Source Reasoning for LLMs (FSR Benchmark):**
 - Developed the FSR benchmark comprising 1,000 multi-source fact bundles to evaluate trust, temporal reasoning, and conflict resolution across 7B–72B LLMs using structured metadata.
 - Investigated how LLMs decide what to trust when sources disagree, identifying a reliance on superficial heuristics and sycophancy over genuine epistemic reasoning (Submitted to AAI).
- **Visual Grounding Benchmark for Vision-Language Models:**
 - Built a counterfactual evaluation framework across 13 VLMs, revealing systematic perceptual bias and identifying visual salience as a key factor governing grounding (Under Review at TMLR).

Resource-Aware AI Systems & Instrumentation

- **Range: Resource-Aware Generation Efficiency (ACM Research):**
 - Introduced an instrumented characterization framework with a usable-yield metric family (UR/J, UR/\$, UR/s) applied to ten models (1.5B–72B).

- Discovered a format-dependent "CSV scale-efficiency penalty," proving that larger models consume disproportionately more energy for structured generation despite higher success rates.
- **Reliability-Resource Tradeoffs in Structured Synthetic Data (SIGMOD Research):**
 - Instrumented a LLaMA-3 8B pipeline with Zeus-ML across 1,980 generation calls to profile multiobjective trade-offs among validity, GPU energy, and latency.
 - Demonstrated that schema enforcement with repair reduces GPU energy consumption by 47.9% and latency by 30.4% relative to unstructured baselines.
- **Multi-Objective Optimization for Renewable Energy Forecasting:**
 - Formulated a MOHPO framework for energy forecasting, jointly optimizing accuracy and hardware energy across ResNet, SVM, and XGBoost using 6 search strategies (e.g., NSGA-III, BoTorch).
 - Instrumented real hardware telemetry via Intel RAPL (CPU) and NVIDIA Zeus (GPU) to decompose energy costs across 24 configurations.
 - Discovered that at matched accuracy, classical models consume up to $226\times$ less energy than deep learning on CPU, but incur a $149\times$ penalty on GPU due to poor algorithmic parallelism.

Reinforcement Learning & Systems Optimization

- **FluxAlloc: Adaptive Dynamic Memory Allocation via RL:**
 - Formulated heap allocation as an MDP, designing a Lookahead+Neural allocator that scores candidate free blocks via n -step simulation rollouts blended with a neural ranker MLP.
 - Achieved a 2.2% absolute reduction in allocation failure rates and 53% lower fragmentation on adversarial workloads by detecting fragmentation-inducing placements.

RESEARCH & ACADEMIC EXPERIENCE

IIIT Bangalore | *Teaching Assistant (AIT 512: Mathematics for Machine Learning)* 2026 – Present

- Instructing graduate students in foundational mathematical concepts for AI, including linear algebra, multivariate calculus, probability, and optimization theory.

NIT Raipur | *Research Intern (BigMint Tech. Collaboration)* Feb 2024 - May 2024

- Designed and developed an end-to-end steel price forecasting pipeline processing 10,000+ daily data points through automated ETL workflows and real-time prediction services.
- Benchmarked 8+ statistical and deep learning forecasting models (ARIMA, XGBoost, LSTM, Transformers) across 12-month prediction horizons.
- Achieved 2.8% MAPE on 500+ test samples and reduced manual forecasting workload by $\approx 40\%$.

IIIT Naya Raipur | *Machine Learning Research Intern* May 2023 - Nov 2023

- Developed an ensemble deep learning framework for multi-class disease classification using multimodal medical imagery, achieving 91.42% accuracy on 9,100+ images.
- Built a non-invasive fitness monitoring platform with IoT sensors and an XGBoost model, achieving 96.67% prediction accuracy; published findings at IEEE SCES.

TECHNICAL SKILLS

Programming: Python, SQL, JavaScript, C++

ML & DL: Deep Learning, NLP, Multimodal Learning, LLMs, VLMs, Foundation Models, Reinforcement Learning, Generative AI.

Frameworks: PyTorch, TensorFlow, Transformers, Scikit-Learn, NumPy, Pandas, Zeus-ML, PyRAPL, BoTorch

Tools: Git, Linux, MLflow, Ollama, Docker, HuggingFace

HONORS & AWARDS

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- **IndiaAI Fellow** (Government of India) for advanced research in Artificial Intelligence and Data Science.
 - **Meta-PyTorch Hackathon 2026** — National Finalist.
 - Led Google Dev Club, MS Student Ambassador, and National Service Scheme (NSS) initiatives.